Prabhavit Gupta

Software Engineer | Customer-Facing AI Agents, Agent Harnesses & Tool Ecosystems

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| Venice, Italy (EU Permanent Resident | Open to Relocation to SF / NY)

Executive Summary

- **Customer-Facing AI Agents & Agent Harnesses:** Software engineer with 3 years commercial software experience engineering production AI agent runtimes, multi-step goal pursuit architectures, and automated customer interaction workflows.
- **Agent Development Life Cycle (ADLC) & Tool Calling:** Experienced across the end-to-end agent life cycle—scoping enterprise business logic, designing tool execution harnesses (Model Context Protocol, REST/GraphQL APIs), and managing continuous prompt and model iteration.
- **Enterprise Reliability & Rigorous Evaluation:** Deep focus on production robustness: constructing systematic evaluation testbeds (LLM-as-judge, DeepEval, Langfuse tracing), context window management, and zero-hallucination tool calling for high-stakes enterprise domains.

Technical Skills & Core Competencies

- **AI Agent Architectures:** Multi-Agent Orchestration, Tool Calling Loops, Model Context Protocol (MCP), Context Engineering, ReAct / Plan-and-Execute, Streaming STT/LLM/TTS Pipelines, Prompt Caching.
- **Evaluation & Observability:** LLM-as-a-Judge, Groundedness & Safety Evals, DeepEval, Langfuse Distributed Tracing, Latency Profiling, Automated Regression Suites.
- **Core Software Development:** TypeScript / JavaScript (Node.js, Next.js, React), Python (AsyncIO, FastAPI, PyTest), RESTful APIs, Webhooks, GraphQL, SQL (PostgreSQL, SQLite), Docker, Git.
- **Deployment & Delivery:** Enterprise System Scoping, Webhook Routing, Third-Party API Integrations, Production Monitoring, Agile Delivery, Open to SF/NY Relocation.

Professional Experience

AI Agent Solutions Engineer & Consultant • *Independent Consultancy*
2024 – Present • Venice, IT (Remote / Hybrid)

- Architected and deployed production customer-facing AI agents for European clients, automating complex multi-step inquiries across booking, subscription management, and user onboarding.
- Implemented Model Context Protocol (MCP) tool servers connecting autonomous agent runtimes with proprietary customer databases and CRM endpoints under strict permission boundaries.
- Designed context engineering pipelines that dynamically retrieve customer history, active entitlements, and real-time state to personalize conversational agent responses.
- Built quantitative eval suites measuring tool selection accuracy, semantic intent resolution, and safety guardrails, reducing unhandled edge cases prior to production release.
- Embedded directly with enterprise operational stakeholders to map multi-turn customer communication journeys into deterministic, auditable agent workflows.

Lead Developer & Product Engineer • *FlowSpace.co*
Mar 2024 – Aug 2026 • London, UK (Remote)

- Owned end-to-end application architecture and customer workflow automation, scaling platform usage across UK and European enterprise hubs to 4,500+ active members.
- Developed and shipped automated conversational support agents that resolved venue access and reservation issues in real time by querying live database state via internal APIs.
- Migrated core monolith services to modern Next.js and backend API microservices, improving page responsiveness and ensuring high operational availability during demand spikes.
- Engineered reliable event-driven webhook connectors with third-party space management APIs (OfficeRnD, Nexudus), automating reservation provisioning with idempotency checks.
- Built automated monitoring and alerting around user communication funnels, rapidly identifying integration errors and maintaining high resolution satisfaction.

Full-Stack Solutions Engineer • *Tech Automation Systems*
2023 – 2024 • Milan / Remote, IT

- Built scalable backend microservices and API integrations in Python and TypeScript connecting client operational interfaces with internal reporting engines.
- Implemented automated unit and end-to-end integration tests within CI/CD pipelines, preventing breaking changes across shared external client interfaces.
- Partnered directly with non-technical business clients to extract domain requirements and deliver clean, testable

software solutions on aggressive timelines.

Key Engineering Projects

- **Autonomous Tool-Calling & Evaluation Harness (TypeScript + MCP + Langfuse):** Architected an enterprise agent runtime enabling LLMs to safely invoke external tool ecosystems with schema-enforced payload validation, full state rollbacks, and distributed tracing.
- **Universal Multi-Source Ingestion Pipeline (Python + Playwright + REST APIs):** Designed an automated data harvesting and enrichment system parsing 900+ structured and semi-structured external sources with schema validation and error isolation.

Education, Languages & Legal Status

- **Bachelor of Arts in Philosophy, International Studies & Economics** – Ca' Foscari University of Venice (2020 – 2024)
  Rigorous foundation in formal logic, epistemology, decision theory, and complex systems analysis.
- **Languages:** English (C1 Full Professional / Fluent), Italian (Working proficiency), Hindi (Native).
- **Legal Status & Availability:** EU Long-Term Resident (Permesso di Soggiorno UE a Tempo Indeterminato, Italy). Open to US relocation to San Francisco or New York with visa sponsorship; available immediately.